

EXPERIENCE LETTER

Date: March 23, 2026

To Whom It May Concern,

This letter is to certify that **Joseph Varghese** (github.com/Josephvarghes | linkedin.com/in/joseph-varghese-ai) engaged with me, **Amaljith K** (github.com/ajk1889 | linkedin.com/in/amaljith-k), as a **Solo Developer** on the project **RestoPulse**, a Gen AI based Restaurant Management and Analytics Platform, under a person-to-person contract spanning approximately **3 months (December 2025 to March 2026)**.

Joseph worked as the sole developer on this project, directly under my technical guidance. He was responsible for the full stack development of RestoPulse, covering backend, frontend, Gen AI integration, and deployment workflows.

Backend Development (Python / FastAPI)

- Designed and implemented RESTful APIs for menu management, order management, authentication, and guest interfaces using **FastAPI** and **SQLAlchemy 2.0 (async)**.
 - Implemented a **multi-tenant JWT authentication system** with refresh token rotation, token blacklisting, and reuse detection.
 - Built **Role Based Access Control (RBAC)** across APIs for management, waiter, kitchen staff, and guest roles.
 - Implemented **PostgreSQL Row Level Security (RLS)** with a three role database architecture to ensure strict tenant data isolation.
 - Worked with **Alembic** for schema migrations and ongoing database evolution.
 - Integrated **Google Cloud Storage (GCS)** with presigned URL based image uploads and an ARQ background task for automated orphan image cleanup.
 - Enabled **PostGIS** based geospatial queries for restaurant location and "near me" style features.
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Gen AI Integration (LangGraph / Google Cloud)

- Built an AI powered dish recommendation system using **LangGraph** agents integrated with **Google Gemini** (via `google.genai`) on **Google Cloud**.
 - Implemented a conversational AI chat interface for guests supporting both text and audio input, including speech to text transcription using GCS.
 - Fine tuned prompts for multilingual support and structured JSON responses, including dish suggestions and customization details.
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Frontend Development (React / Vite / TailwindCSS)

- Developed the **Restaurant Management Interface** and **Guest Interface** in a unified monorepo using **React**, **Vite**, **TailwindCSS v4**, and **Radix UI** components.
 - Implemented real time cart management, order tracking, and AI chat UI with markdown rendering and audio recording support on the guest side.
 - Integrated **Firebase Cloud Messaging (FCM)** for cross device notifications such as waiter call alerts, kitchen order notifications, and acknowledgment flows.
 - Implemented a PDF export feature for order reports with currency formatting and date range handling for management users.
 - Managed frontend authentication using **axios-jwt**, including automatic token refresh, 401 handling, and centralized state via **Zustand**.
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DevOps and Infrastructure

- Configured a complete **Docker Compose** stack (FastAPI backend, React/Vite frontend, Nginx, PostgreSQL 17, Redis 7) for local development and production deployment.
 - Set up wildcard **Let's Encrypt** SSL certificates and Nginx reverse proxy for multi tenant subdomain routing on Ubuntu virtual machines.
 - Established code quality automation using **pre-commit**, **Ruff**, **Husky**, and deterministic dependency management with **uv**.
 - Wrote and maintained **Pytest** based test suites with async isolation, per test database transaction rollback, and RLS context simulation.
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During this engagement, Joseph consistently demonstrated strong ownership of the codebase, from schema design and API contracts to frontend UX and production deployment details. He handled complex technical challenges such as race condition resolution, transactional integrity, concurrent notification delivery, and multi tenant security, largely independently with architectural guidance from me.

I am pleased to recommend **Joseph Varghese** for any full stack or backend engineering role, especially those involving Python, FastAPI, React, Gen AI systems, and cloud native architectures.

Sincerely,

Amaljith K
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